Lavneet Hora

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Education

Texas Tech University — **Expected Graduation: May 2027**

Bachelor of Science in Computer Science (Honors), Minor in Math and Bioengineering — *GPA: 3.94*

- Awards: 4x President's Honor List, IEEE Computer Society Career Catalyst Scholar
- Leadership & Activities: Officer @ Google Developers Group, Hacker Experience Lead @ HackWesTX 2025

Experience

UTHealth Houston, BIG-TCR Program — **June 2026 – August 2026**

Research Intern — *Houston, TX*

- Built an R/Seurat + AUCell single-cell RNA-seq pipeline that scores 60,000 PDAC tumor cells for neural-pathway activity and clusters them into three subtypes, extending the lab's bulk findings to single-cell resolution.

Texas Tech University, Dept. of Electrical and Computer Engineering — **May 2025 – October 2025**

Undergraduate Research Assistant — *Lubbock, TX*

- Identified 5 key behavioral indicators for anxiety prediction by analyzing movement patterns across 100+ patient datasets using statistical correlation analysis.
- Reduced preprocessing time by 50% by implementing automated MATLAB pipelines for DFA/MF-DFA analysis and Hurst exponent visualization.
- Standardized trend analysis methodology across 100+ daily movement profiles, improving reproducibility and reducing analysis variance by 30%.

iMocha — **July 2025 – August 2025**

GenAI Intern — *Remote*

- Reduced QA triage time by 75% (12→3 min/report) by building Gemini API classification pipeline that automated issue categorization for 500+ monthly coding assessments.
- Decreased resolution cycles by 40% by engineering Python ETL workflows that transformed 1,000+ candidate feedback entries into structured QA reports.

Projects

Ceremoni | *FastAPI, Celery, Redis, Azure TTS, Microsoft OAuth, Docker* — **Demo**

- Built an end-to-end AI graduation name pronunciation system that records student self-pronunciations, extracts IPA phonetic notation, and generates high-quality audio, ensuring every name is announced correctly regardless of cultural origin.
- Engineered full backend pipeline with FastAPI, Celery task queues, Redis, and Azure TTS, containerized with Docker, with Microsoft OAuth for secure institutional login.
- Designed for institutional deployment at scale, capable of serving 5,000+ graduating students annually across any ceremony size with automated name processing and live playback controls.

RaiderRating | *TypeScript, React, Plasmo, Express, Node.js, Railway* — **Demo**

- Built a Chrome extension that shows Rate My Professors ratings inside TTU's Schedule Builder, saving students from tab-switching during registration - 175+ installs and a 5.0/5.0 rating on the Chrome Web Store.
- Engineered full-stack architecture with Plasmo and React frontend, Express/Node.js backend deployed on Railway, enabling real-time professor data fetching with zero user configuration.
- Eliminated manual professor lookup by automating name matching and overlaying rating badges showing overall score, difficulty, and "would take again" percentage directly in the schedule builder.

Football Kick Analyzer | *Rubik Pi 3, Python, YOLOv8, OpenCV, MediaPipe* — **Demo**

- Built a real-time AI system for football players to analyze kick accuracy and trajectory by detecting and tracking ball flight on embedded hardware - trained a custom YOLOv8 model on a self-collected dataset of 3,000+ images achieving 20-35ms inference latency on Rubik Pi 3.
- Engineered smooth trajectory reconstruction using Kalman filtering and spline interpolation to handle noisy detections across motion blur and varied lighting conditions.
- Presented live demo to Qualcomm engineers and leadership team during IEEE Career Catalyst headquarters visit.

Technical Skills

Languages: Python, Java, C++, C, SQL, MATLAB, JavaScript, R

Frameworks & Libraries: React, PyTorch, OpenCV, TensorFlow, Next.js, Node.js, FastAPI, Tailwind CSS, Pandas, NumPy, Qiskit, Selenium

Tools & Platforms: Git, Docker, Firebase, Supabase, Jupyter, VS Code, Google Cloud, Linux